Gimbal Mount Assembly
Requirement Consolidation

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1. Introduction

1.1. Scope

Based on the preliminary requirements initially shared by The Exploration Company (TEC) for this activity [RD01], this document refines the key performance parameters. These updated requirements primarily relate to the Hopper Ground Demonstrator *Oneiros*, which is also part of a de-risking activity for the *Huracan* propulsive system [RD02]. During the development of the Gimbal Mount Assembly (GMA), the focus will be on prioritizing *Oneiros* requirements, while also considering intentions for upcoming design iterations of the *Huracan* engine and its components.

1.2. Reference

[RD01] Consultant Agreement / AA5722EA-0E20-45A6-BFE2-2B6D313A7FF3 / 2025

[RD02] Design Description and Justification File / TEC-ITA-DOC-2025-01011 / Version 3

[RD03] System Requirements Document / TEC-ITA-DOC-2025-01017 / Version 0

[RD04] Modern Engineering For Design Of Liquid Propelleant Rocket Engines - D.Huzel / 1992

2. Requirements

2.1. Environmental requirements

REQ-033 - Pollution requirements

Sensitive areas, such as regions of relative motion, shall be protected against sand particles with an average size of 0.2 mm at a temperature of 49.6 °C [RD03].

REQ-020 - Ambient requirements

AccoRD0ing to the system requirements of *Oneiros*, the sea level conditions are defined by a pressure of 0.994 bar to 1.0276 bar and 12 % to 91 % humidity. The ambient temperature ranges from 8.3 °C to 49.6 °C. [RD03]

2.2. Functional requirements

REQ-003 - Operational gimbal capability

The GMA design provides an operational deflection capability of ±10° per axis (pitch and yaw), which covers the GNC needs as well as the mechanical limits of the subsystem and its components (e.g., bellows and actuators).

REQ-034 - Max. gimbal capability

For off-nominal cases and emergency authority, the max. gimbal angle shall not exceed +/-12 °.

REQ-002 - Rotation axis

To provide full pitch and yaw control capability, a two-axis rotation mechanism shall be implemented. Both axes are positioned perpendicular to each other within the same horizontal plane.

REQ-004 - Angular velocity

To compromise GNC needs with mechanical constraints of the engine such as the actuator capacity, the max. angular velocity of the engine is 20 °/s.

REQ-005 - Angular acceleration

The engine´s acceleration and hence, the capability of the GMA shall be max. 25 rad/s^2.

REQ-0014 - Mass

The maximum mass of the GMA shall be equal or below 5 kg with a measurement accuracy of +/-0.1 %.

REQ-013 - Geometrical envelope

A damage- and collision-free motion under worst case angle conditions is required between the Igniter´s main body (incl. it´s attachments) and the GMA geometry. A minimum clearance of 20 mm is to be considered between moving parts. The maximum geometrical limit of th GMA is given by the mass constraint, the loads and (thermo-)mechanical stress requirements mentioned in this document. The Ignition System´s main body (Ø70 mm) is designed with a vertically offset of 63 mm from the Injection Head (IH) upper flange surface.

REQ-015 - Geometrical interface

For the lower attachment of the GMA, the utilizable surface of the IH features a ring with an inner diameter of Ø100 mm and an outer diameter of Ø170 mm. In total, eight through-holes (Ø9 mm each) are available, spaced at 45° intervals (symetrically along PCD). Eight M8 fasteners are intended to fix the GMA with the IH (minimum tensile ultimate strength per fastener: 800 MPa).

2.3. Mechanical and thermal requirements

REQ-016 - Max. payload

The gimbal will be designed to support a maximum payload mass of 40 kg, corresponding to the Thrust Chamber Assembly (TCA) plus margins of the cryogenic engine.

REQ-018 - Friction coefficient

To reduce the breaking torque of the actuators, the GMA shall provide a constant friction coefficient equal or lower than 0.1 under mechanical and thermal operating conditions.

REQ-008 - Plastic Deformation and crack formation

The GMA and its parts shall not undergo plastic deformation, crack formation or other non-recoverable deformation when subjected to the operational conditions defined in this specification. Exceptions can be permissible if non-recoverable deformation is identified as a single, local phenomena, that occurs in extraoRD0inary or single event load case. In this case the non-recoverable deformation is to be justified by analysis at least.

REQ-001 - Operating nominal thrust

The GMA shall transmit a total nominal thrust load of 15 kN.

REQ-039 - Quasi-Static Load (QSL)

For single events, the GMA shall sustain an excessive thrust load of 22.5 kN at gimbal angle of 0 °, which includes a safety factor of 1.5 applied to the nominal thrust.

REQ-021 - Mechanical and thermal transient

Until steady state conditions are achieved, transient conditions dominate. A cooling from ambient 280 K to 180 K within 10 s is to be considered linearly, with a slope of 10 K/s. This is to be taken into account at the GMA surface, that is connected to the IH upper flange. A chamber overpressure rate of 210 bar/s is to be anticipated for a transinent occasion.

REQ-020 - Operating temperature

An operating temperature of 120 K to 150 K shall be considered at the IH top flange surface that is connected to the GMA support.

REQ-030 - Cycling

The GMA shall sustain a minimum of 1000 cycles, which is derived from preliminary GNC-data.

2.4. Growth potential requirements

REQ-035 - Extended cycling

For further utilization of the GMA for flight qualification and preperation for the moon mission, 26520 cycles are targeted.

REQ-036 - Lateral misalignment

The GMA shall contain features to compensate lateral misalignment of the thrust vector up to +/-10 mm [RD04].

REQ-037 - Vacuum environment

The GMA performance in ambient conditions shall be adapted to operate feasibly in vacuum, especially concerning lubrication, thermal and mechanical loads.

3. Acronym List

The acronyms used in this document are listed below.

Acronym	Definition
IH	Injection Head flange
GMA	Gimbal Mount Assembly
PCD	Pitch Circle diameter
QSL	Quasi-static load
TEC	The Exploration Company